

Assignment 01 - Limits

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Part 4: Practice Assignments

1.

$$\lim_{x \rightarrow 4} (x + 6)$$

2.

$$\lim_{x \rightarrow 3} (4x)$$

3.

$$\lim_{x \rightarrow 2} (x^2 + 1)$$

4.

$$\lim_{x \rightarrow 10} \left(\frac{x + 2}{3} \right)$$

5.

$$\lim_{x \rightarrow 9} \sqrt{x}$$

6.

$$\lim_{x \rightarrow 0} (3x^2 + 8x - 2)$$

7.

$$\lim_{x \rightarrow 100} 15$$

8.

$$\lim_{x \rightarrow 4} \frac{x^2 - 16}{x - 4}$$

(Hint: Use the Difference of Squares rule!)

9.

$$\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1}$$

(Hint: 1 is a perfect square because $1 \times 1 = 1$)

10.

$$\lim_{x \rightarrow 3} \frac{4x - 12}{x - 3}$$

(Hint: Pull out the Greatest Common Factor from the top!)

11. The Rocket Launch

A model rocket is launched, and its altitude in feet is measured by the equation $Altitude = x^2$ (where x is seconds).

Set up a limit and use the factoring trick to find the rocket's exact speed at exactly **4 seconds**.

(Hint: Set up the limit as $x \rightarrow 4$, use x^2 for the moving distance and 4^2 for the fixed distance).
